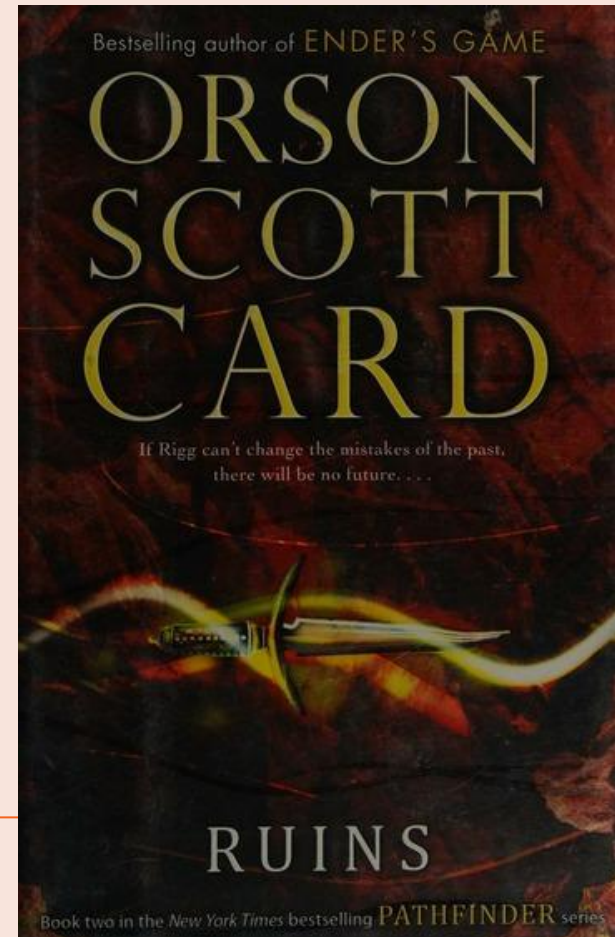


# A Foray into Clockmaking

Cutler Dunn

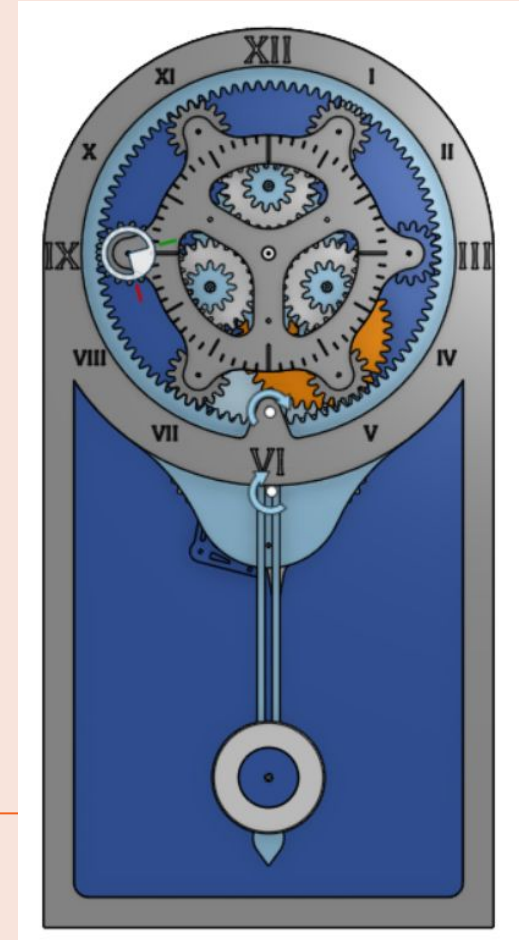
Of course you can travel  
to the future - at the rate  
of one second per  
second, just like  
everybody else.

-Orson Scott Card, *Ruins*



# Key Issues to Consider

1. Escapement design
2. Gear train manufacturing and materials
3. Power delivery

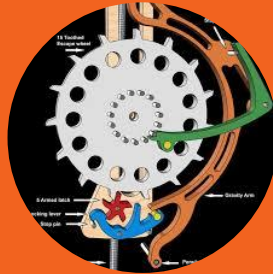


# Escapement Designs



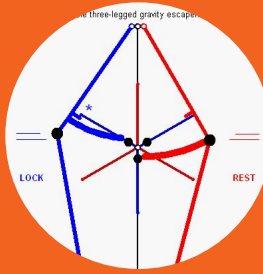
**Arnfield Escapement**

A robust escapement designed to provide consistent power to the pendulum while isolating the pendulum from powertrain variation



**Modified Arnfield  
(Law Wooden Clocks)**

Adopts the isolation of the pendulum from the powertrain, but takes it a step further to decrease friction



**Double Three-legged  
Escapement**

A clever escapement used in the huge pendulum clocks at Trinity college and Big Ben in the UK



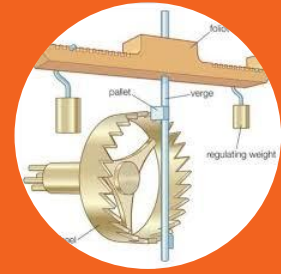
**Grasshopper  
Escapement**

An odd-looking and complex but effective escapement, uses two legs in a push-and-pull arrangement to



**Recoil Anchor  
Escapement**

Possibly the simplest escapement mechanism, relies on the reversal of the powertrain at the beginning of the swing to provide power to the pendulum.



**Verge and Foliot  
Escapement**

Appears in some of the oldest mechanical clocks, possibly the first strictly mechanical clock design used it. Relies on swinging weights rather than a pendulum.

# Gear Manufacture

## 1. 3D Printing

- A. Selective Laser Sintering(SLS): provides good material properties and faithful part production
- B. Fused Deposition Modeling (FDM): may have trouble with small features and layering, accessible
- C. Stereolithography(SLA): produces accurate details, can be expensive

## 2. Laser Cutting or CNC Machining

Can produce good parts but may require programming ability.  
In my experience, laser cutting gears removes part of the profile and causes meshing problems.

## 3. Gear cutting

This method is traditional for clocks, as well as the best way to get accurate gears. However, it is very hard to do right manually and can be very expensive.

# Gear Materials

## 1. Wood

It can be done, and it looks very pretty, but getting faithful, consistent gear teeth is notoriously difficult.

## 2. Metal

Difficult and expensive to machine, but highly wear resistant and capable of having small features without breaking

## 3. Plastics

Generally easy to manufacture, and usually good for faithful part reproduction. Some plastics can be found that are wear resistant and self-lubricating.

Design requirements

What makes a design useful & beginner friendly?

01

Price

02

Ease & Versatility of  
Manufacture

03

Simple to Understand  
Design

Any questions?  
Ask away!



## Image References

Slide 2:

Cover of Ruins, by Orson Scott Card, found on Internet Archive:

[https://www.google.com/url?sa=i&url=https%3A%2F%2Farchive.org%2Fdetails%2Fruins0000card\\_q6y6&psig=AOvVaw2dfpJHwwdhxF17qVEXX-1Q&ust=1763228506493000&source=images&cd=vfe&opi=89978449&ved=0CBkQjhxqFwoTCMCfoZmY8pADFQAAAAAdAAAAABAL](https://www.google.com/url?sa=i&url=https%3A%2F%2Farchive.org%2Fdetails%2Fruins0000card_q6y6&psig=AOvVaw2dfpJHwwdhxF17qVEXX-1Q&ust=1763228506493000&source=images&cd=vfe&opi=89978449&ved=0CBkQjhxqFwoTCMCfoZmY8pADFQAAAAAdAAAAABAL)

Slide 4:

Arnfield Escapement on my-time-machines:

[https://www.my-time-machines.net/croce\\_arnfield.htm](https://www.my-time-machines.net/croce_arnfield.htm)

Wooden Clocks: Designing a new Gravity Escapement by Brian Law:

<https://brianlawswoodenclocks.blogspot.com/2014/02/designing-new-gravity-escapement.html>

The Trinity Clock on the Trinity College website:

[https://clock.trin.cam.ac.uk/main.php?menu\\_option=escapement](https://clock.trin.cam.ac.uk/main.php?menu_option=escapement)

The Grasshopper Escapement on Engineezy:

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Anchor Escapement on Wikipedia:

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Verge and Foliot Escapement:

<https://www.cabinet.ox.ac.uk/key-innovation-escapement>